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PRACTICE PROBLEMS

1. The nominal risk-free rate is *best* described as the sum of the real risk-free rate and a premium for:

   A. maturity.

   B. liquidity.

   C. expected inflation.

2. Which of the following risk premiums is most relevant in explaining the difference in yields between 30-year bonds issued by the US Treasury and 30-year bonds issued by a small, private US corporate issuer?

   A. Inflation

   B. Maturity

   C. Liquidity

3. Consider the following annual return for Fund Y over the past five years:

Exhibit 1: Five-Year Annual Returns

Year	Return (%)
Year 1	19.5
Year 2	–1.9
Year 3	19.7
Year 4	35.0
Year 5	5.7

   The geometric mean return for Fund Y is *closest* to:

   A. 14.9 percent.

   B. 15.6 percent.

   C. 19.5 percent.

4. A portfolio manager invests EUR5,000 annually in a security for four years at the following prices:

Exhibit 1: Five-Year Purchase Prices

Year	Purchase Price (euros per unit)
Year 1	62.00
Year 2	76.00

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Year	Purchase Price (euros per unit)
Year 3	84.00
Year 4	90.00

The average price is *best* represented as the:

A. harmonic mean of EUR76.48.

B. geometric mean of EUR77.26.

C. arithmetic average of EUR78.00.

5. Which of the following statements regarding arithmetic and geometric means is correct?

A. The geometric mean will exceed the arithmetic mean for a series with non-zero variance.

B. The geometric mean measures an investment's compound rate of growth over multiple periods.

C. The arithmetic mean measures an investment's terminal value over multiple periods.

6. A fund receives investments at the beginning of each year and generates returns for three years as follows:

Exhibit 1: Investments and Returns for Three Years

Year of Investment	Assets under Management at the Beginning of each year	Return during Year of Investment
1	USD1,000	15%
2	USD4,000	14%
3	USD45,000	–4%

Which return measure over the three-year period is negative?

A. Geometric mean return

B. Time-weighted rate of return

C. Money-weighted rate of return

7. At the beginning of Year 1, a fund has USD10 million under management; it earns a return of 14 percent for the year. The fund attracts another net USD100 million at the start of Year 2 and earns a return of 8 percent for that year. The money-weighted rate of return of the fund is *most likely* to be:

A. less than the time-weighted rate of return.

B. the same as the time-weighted rate of return.

C. greater than the time-weighted rate of return.

8. An investor is evaluating the returns of three recently formed ETFs. Selected

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return information on the ETFs is presented in Exhibit 1:

Exhibit 1: Returns on ETFs

ETF	Time Since Inception	Return Since Inception (%)
1	125 days	4.25
2	8 weeks	1.95
3	16 months	17.18

Which ETF has the highest annualized rate of return?

A. ETF 1

B. ETF 2

C. ETF 3

9. The price of a stock at $t = 0$ is USD208.25 and at $t = 1$ is USD186.75. The continuously compounded rate of return, $r_{1,T}$ for the stock from $t = 0$ to $t = 1$ is *closest* to:

A. –10.90 percent.

B. –10.32 percent.

C. 11.51 percent.

10. A USD25 million equity portfolio is financed 20 percent with debt at a cost of 6 percent annual cost. If that equity portfolio generates a 10 percent annual total investment return, then the leveraged return is:

A. 11.0 percent.

B. 11.2 percent.

C. 13.2 percent

11. An investment manager's gross return is:

A. an after-tax nominal, risk-adjusted return.

B. the return earned by the manager prior to deduction of trading expenses.

C. an often used measure of an investment manager's skill because it does not include expenses related to management or administration.

12. The strategy of using leverage to enhance investment returns:

A. amplifies gains but not losses.

B. doubles the net return if half of the invested capital is borrowed.

C. increases total investment return only if the return earned exceeds the borrowing cost.

13. At the beginning of the year, an investor holds EUR10,000 in a hedge fund. The investor borrowed 25 percent of the purchase price, EUR2,500, at an annual interest rate of 6 percent and expects to pay a 30 percent tax on the return she

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earns from his investment. At the end of the year, the hedge fund reported the information in Exhibit 1:

Exhibit 1: Hedge Fund Investment

Gross return	8.46%
Trading expenses	1.10%
Managerial and administrative expenses	1.60%

The investor's after-tax return on the hedge fund investment is closest to:

A. 3.60 percent.

B. 3.98 percent.

C. 5.00 percent.

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SOLUTIONS

1. C is correct. The nominal risk-free rate is approximated as the sum of the real risk-free interest rate and an inflation premium.

2. C is correct. US Treasury bonds are highly liquid, whereas the bonds of small issuers trade infrequently and the interest rate includes a liquidity premium. This liquidity premium reflects the relatively high costs (including the impact on price) of selling a position. As the two bond issues have the same 30-year maturity, the observed difference in yields would not be solely explained by maturity. Further, the inflation premium embedded in the yield of both bonds is likely to be similar given they are both US-based bonds with the same maturity.

3. A is correct. The geometric mean return for Fund Y is calculated as follows:

   $\overline{R}_G = [(1 + 0.195) \times (1 - 0.019) \times (1 + 0.197) \times (1 + 0.350) \times (1 + 0.057)]^{(1/5)} - 1 = 14.9\%$.

4. A is correct. The harmonic mean is appropriate for determining the average price per unit as it gives equal weight to each data point and reduces the potential influence of outliers. It is calculated as follows:

   $\overline{X}_H = 4/[(1/62.00) + (1/76.00) + (1/84.00) + (1/90.00)] = \text{EUR}76.48$.

5. B is correct. The geometric mean compounds the periodic returns of every period, giving the investor a more accurate measure of the terminal value of an investment.

6. C is correct. The money-weighted rate of return considers both the timing and amounts of investments into the fund. To calculate the money-weighted rate of return, tabulate the annual returns and investment amounts to determine the cash flows.

Year	1	2	3
Balance from previous year	0	USD1,150	USD4,560
New investment	USD1,000	USD2,850	USD40,440
Net balance at the beginning of year	USD1,000	USD4,000	USD45,000
Investment return for the year	15%	14%	–4%
Investment gain (loss)	USD150	USD560	–USD1,800
Balance at the end of year	USD1,150	USD4,560	USD43,200

$CF_0 = -\text{USD}1{,}000$, $CF_1 = -\text{USD}2{,}850$, $CF_2 = -\text{USD}40{,}440$, $CF_3 = +\text{USD}43{,}200$.

$$CF_0 = -1{,}000$$
$$CF_1 = -2{,}850$$
$$CF_2 = -40{,}440$$
$$CF_3 = +43{,}200$$

$$\frac{CF_0}{(1+\text{IRR})^0} + \frac{CF_1}{(1+\text{IRR})^1} + \frac{CF_2}{(1+\text{IRR})^2} + \frac{CF_3}{(1+IRR)^3}$$
$$= \frac{-1{,}000}{1} + \frac{-2{,}850}{(1+\text{IRR})^1} + \frac{-40{,}440}{(1+\text{IRR})^2} + \frac{43{,}200}{(1+\text{IRR})^3} = 0$$

Solving for *IRR* results in a value of *IRR* = –2.22 percent.

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Note that A and B are incorrect because the time-weighted rate of return (TWR) of the fund is the same as the geometric mean return of the fund and is positive:

$$R_{TW} = \sqrt[3]{(1.15)(1.14)(0.96)} - 1 = 7.97\%.$$

7. A is correct. Computation of the money-weighted return, *r*, requires finding the discount rate that sums the present value of cash flows to zero. Because most of the investment came during Year 2, the money-weighted return will be biased toward the performance of Year 2 when the return was lower. The cash flows are as follows:

$$CF_0 = -10$$

$$CF_1 = -100$$

$$CF_2 = +120.31$$

The terminal value is determined by summing the investment returns for each period [(10 × 1.14 × 1.08) + (100 × 1.08)].

$$\frac{CF_0}{(1+IRR)^0} + \frac{CF_1}{(1+IRR)^1} + \frac{CF_2}{(1+IRR)^2}$$
$$\frac{-10}{1} + \frac{-100}{(1+IRR)^1} + \frac{120.31}{(1+IRR)^2} = 0.$$

This results in a value of *IRR* = 8.53 percent.

The time-weighted return of the fund is calculated as follows:

$$R_{TW} = \sqrt[2]{(1.14)(1.08)} - 1 = 10.96\%.$$

8. B is correct. The annualized rate of return for

ETF 1 annualized return = $(1.0425^{365/125}) - 1 = 12.92\%$

ETF 2 annualized return = $(1.0195^{52/8}) - 1 = 13.37\%$

ETF 3 annualized return = $(1.1718^{12/16}) - 1 = 12.63\%$

Despite having the lowest value for the periodic rate, ETF 2 has the highest annualized rate of return because of the reinvestment rate assumption and the compounding of the periodic rate.

9. A is correct. The continuously compounded return from $t = 0$ to $t = 1$ is $r_{0,1} = \ln(S_1/S_0) = \ln(186.75/208.25) = -0.10897 = -10.90\%$.

10. A is correct.

$$R_L = R_p + \frac{V_B}{V_E}(R_p - r_D)$$

$$= 10\% + \frac{USD5\ million}{USD20\ million}(10\% - 6\%)$$

$$= 10\% + 0.25 \times 4\% = 11.0\%.$$

11. C is correct. Gross returns are calculated on a pre-tax basis; trading expenses *are* accounted for in the computation of gross returns as they contribute directly to the returns earned by the manager. A is incorrect because investment managers' gross returns are pre-tax and not adjusted for risk. B is incorrect because managers' gross returns do reflect the deduction of trading expenses since they contribute directly to the return earned by the manager.

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12. C is correct. The use of leverage can increase an investor's return if the total investment return earned on the leveraged investment exceeds the borrowing cost on debt. A is incorrect because leverage amplifies both gains and losses. B is incorrect because, if half of the invested capital is borrowed, then the investor's gross (not net) return would double.

13. C is correct. The first step is to compute the investor's net return from the hedge fund investment. The net return is the fund's gross return less managerial and administrative expenses of 1.60 percent, or 8.46% – 1.60% = 6.86%. Note that trading expenses are already reflected in the gross return, so they are not subtracted.
The second step is to compute the investor's leveraged return (the investor borrowed EUR2,500 (25 percent) of the purchase), calculated as: follows

$$R_L = R_p + \frac{V_B}{V_E}(R_p - r_D)$$

$$R_L = 6.86\% + \frac{EUR2,500}{EUR7,500}(6.86\% - 6\%)$$

$R_L = 6.86\% + 0.33 \times 0.86\% = 7.15\%$.

The final step is to compute the after-tax return:
After-tax return = 7.15% (1 – 0.30) = 5.00%.

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PRACTICE PROBLEMS

1. Grupo Ignacia issued 10-year corporate bonds two years ago. The bonds pay an annualized coupon of 10.7 percent on a semiannual basis, and the current annualized YTM is 11.6 percent. The current price of Grupo Ignacia's bonds (per MXN100 of par value) is *closest* to:

   A. MXN95.47.

   B. MXN97.18.

   C. MXN95.39.

2. Grey Pebble Real Estate seeks a fully amortizing fixed-rate five-year mortgage loan to finance 75 percent of the purchase price of a residential building that costs NZD5 million. The annual mortgage rate is 4.8 percent. The monthly payment for this mortgage would be *closest* to:

   A. NZD70,424.

   B. NZD93,899.

   C. NZD71,781.

3. Mylandia Corporation pays an annual dividend to its shareholders, and its most recent payment was CAD2.40. Analysts following Mylandia expect the company's dividend to grow at a constant rate of 3 percent per year in perpetuity. Mylandia shareholders require a return of 8 percent per year. The expected share price of Mylandia is *closest* to:

   A. CAD48.00.

   B. CAD49.44.

   C. CAD51.84.

4. Suppose Mylandia announces that it expects significant cash flow growth over the next three years, and now plans to increase its recent CAD2.40 dividend by 10 percent in each of the next three years. Following the 10 percent growth period, Mylandia is expected to grow its annual dividend by a constant 3 percent indefinitely. Mylandia's required return is 8 percent. Based upon these revised expectations, The expected share price of Mylandia stock is:

   A. CAD49.98.

   B. CAD55.84.

   C. CAD59.71.

5. Consider a Swiss Confederation zero-coupon bond with a par value of CHF100, a remaining time to maturity of 12 years and a price of CHF89. In three years' time, the bond is expected to have a price of CHF95.25. If purchased today, the bond's expected annualized return is *closest* to:

   A. 0.58 percent.

   B. 1.64 percent.

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C. 2.29 percent.

6. Grupo Ignacia issued 10-year corporate bonds four years ago. The bonds pay an annualized coupon of 10.7 percent on a semiannual basis, and the current price of the bonds is MXN97.50 per MXN100 of par value. The YTM of the bonds is *closest* to:

A. 11.28 percent.

B. 11.50 percent.

C. 11.71 percent.

7. Mylandia Corporation stock trades at CAD60.00. The company pays an annual dividend to its shareholders, and its most recent payment of CAD2.40 occurred yesterday. Analysts following Mylandia expect the company's dividend to grow at a constant rate of 3 percent per year. Mylandia's required return is:

A. 8.00 percent.

B. 7.00 percent.

C. 7.12 percent.

8. An analyst observes the benchmark Indian NIFTY 50 stock index trading at a forward price-to-earnings ratio of 15. The index's expected dividend payout ratio in the next year is 50 percent, and the index's required return is 7.50 percent. If the analyst believes that the NIFTY 50 index dividends will grow at a constant rate of 4.50 percent in the future, which of the following statements is correct?

A. The analyst should view the NIFTY 50 as overpriced.

B. The analyst should view the NIFTY 50 as underpriced.

C. The analyst should view the NIFTY 50 as fairly priced.

9. If you require an 8 percent return and must invest USD500,000, which of the investment opportunities in Exhibit 1 should you prefer?

Exhibit 1: Investment Opportunities

Cash flows (in thousands)	t = 0	t = 1	t = 2	t = 3
Opportunity 1	–500	195	195	195
Opportunity 2	–500	225	195	160.008

A. Opportunity 1

B. Opportunity 2

C. Indifferent between the two opportunities.

10. Italian one-year government debt has an interest rate of 0.73 percent; Italian two-year government debt has an interest rate of 1.29 percent. The breakeven one-year reinvestment rate, one year from now is *closest* to:

A. 1.01 percent.

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B. 1.11 percent.

C. 1.85 percent.

11. The current exchange rate between the euro and US dollar is USD/EUR1.025. Risk-free interest rates for one year are 0.75 percent for the euro and 3.25 percent for the US dollar. The one-year USD/EUR forward rate that *best* prevents arbitrage opportunities is:

A. USD/EUR1.051.

B. USD/EUR1.025.

C. USD/EUR0.975.

12. A stock currently trades at USD25. In one year, it will either increase in value to USD35 or decrease to USD15. An investor sells a call option on the stock, granting the buyer the right, but not the obligation, to buy the stock at USD25 in one year. At the same time, the investor buys 0.5 units of the stock. Which of the following statements about the value of the investor's portfolio at the end of one year is correct?

A. The portfolio has a value of USD7.50 in both scenarios.

B. The portfolio has a value of USD25 in both scenarios.

C. The portfolio has a value of USD17.50 if the stock goes up and USD7.50 if the stock goes down.

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SOLUTIONS

1. C is correct. The coupon payments are 5.35 (=10.7/2), the discount rate is 5.8 percent (=11.6%/2) per period, and the number of periods is 16 (=8×2). Using Equation 6, the calculation is as follows:

$$95.39 = \frac{5.35}{1.058} + \frac{5.35}{1.058^2} + \frac{5.35}{1.058^3} + \frac{5.35}{1.058^4} + \ldots + \frac{5.35}{1.058^{14}} + \frac{5.35}{1.058^{15}} + \frac{105.35}{1.058^{16}}.$$

Alternatively, using the Microsoft Excel or Google Sheets PV function (`PV(0.058,16,5.35,100,0)`) also yields a result of MXN95.39.

A is incorrect. MXN95.47 is the result when incorrectly using coupon payments of 10.7, a discount rate of 11.6 percent, and 8 as the number of periods.

B is incorrect. MXN97.18 is the result when using the correct semiannual coupons and discount rate, but incorrectly using 8 as the number of periods.

2. A is correct. The present value of the mortgage is NZD3.75 million (=0.75×5,000,000), the periodic discount rate is 0.004 (=0.048/12), and the number of periods is 60 (=5×12). Using Equation 8,

$$A = \$70,424 = \frac{0.4\%\ (\text{NZD}3,750,000)}{1 - (1 + 0.4\%)^{-60}}.$$

Alternatively, the spreadsheet PMT function may be used with the inputs stated earlier.

B is incorrect. NZD93,899 is the result if NZD5 million is incorrectly used as the present value term.

C is incorrect. NZD71,781 is the result if the calculation is made using 4.8 percent as the rate and 5 as the number of periods, then the answer is divided by 12.

3. B is correct. Mylandia's next expected dividend is CAD2.472 (=2.40×1.03), and using Equation 14,

$$PV_t = \frac{2.40(1 + 0.03)}{0.08 - 0.03} = \frac{2.472}{0.05} = 49.44.$$

4. C is correct. Following the first step, we observe the following expected dividends for Mylandia for the next three years:

In 1 year: D_1 = CAD2.64 (=2.40×1.10)

In 2 years: D_2 = CAD2.90 (=2.40×1.10^2)

In 3 years: D_3 = CAD3.19 (=2.40×1.10^3)

The second step involves a lower 3 percent growth rate. At the end of year four, Mylandia's dividend (D_4) is expected to be CAD3.29 (=2.40×1.10^3×1.03). At this time, Mylandia's expected terminal value at the end of three years is CAD65.80 using Equation 17, as follows:

$$E(S_{t+n}) = \frac{3.29}{0.08 - 0.03} = 65.80.$$

Third, we calculate the sum of the present values of these expected dividends using Equation 16:

$$PV_t = \frac{2.64}{1.08} + \frac{2.90}{1.08^2} + \frac{3.19}{1.08^3} + \frac{65.80}{1.08^3} = 59.71.$$

5. C is correct. The FV of the bond is CHF95.25, the PV is CHF89, and the number of annual periods (t) is 3. Using Equation 18,

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2.29 percent = (92.25/89)(1/3) − 1.

A is incorrect as the result is derived using t of 12. B is incorrect as this result is derived using a PV of CHF95.25 and an FV of 100.

6. A is correct. The YTM is calculated by solving for the RATE spreadsheet function with the following inputs: number of periods of 12 (=6 × 2), coupon payments of 5.35 (=10.7/2), PV of –97.50, and FV of 100. The resulting solution for RATE of 5.64 percent is in semiannual terms, so multiply by 2 to calculate annualized YTM of 11.28 percent.

   B is incorrect, as 11.50 percent is the result if number of periods used is eight, instead of 12. C is incorrect, as 11.71 percent is the result if the number of periods used is 6, instead of 12.

7. C is correct. We may solve for required return based upon the assumption of constant dividend growth using Equation 21:

$$r = \frac{2.40(1.03)}{60} + 0.03 = 0.0712.$$

   B is incorrect as 7.00 percent is the result if we use the previous dividend of CAD2.40 instead of the next expected dividend. A is incorrect as 8.00 percent is simply the required return assumed from one of the Mylandia examples in Question Set 1 in which the price is solved to be a lower value.

8. B is correct. Using Equation 24, the previous input results in the following inequality:

$$15 < \frac{0.50}{0.075 - 0.045} = 16.67.$$

   The above inequality implies that the analyst should view the NIFTY 50 as priced too low. The fundamental inputs into the equation imply a forward price to earnings ratio of 16.67 rather than 15. An alternative approach to answering the question would be to solve for implied growth using the observed forward price to earnings ratio of 15 and compare this to the analyst's growth expectations:

$$15 = \frac{0.50}{0.075 - g}.$$

   Solving for g yields a result of 4.1667 percent. Since the analyst expects higher NIFTY 50 dividend growth of 4.50 percent, the index is viewed as underpriced.

9. C is correct. Using cash flow additivity, compare the two opportunities by subtracting Opportunity 2 from Opportunity 1 yielding the following cash flows:

Opportunity 1 – Opportunity 2	0	–30	0	34.992

   Finding the present value of the above cash flows at 8 percent discount rate shows that both investment opportunities have the same present value. Thus, the two opportunities are economically identical, and there is no clear preference for one over the other.

10. C is correct. The one-year forward rate reflects the breakeven one-year reinvestment rate in one year, computed as follows:

$$F_{1,1} = (1+r_2)^2/(1+r_1) - 1,$$

$$F_{1,1} = (1.0129)^2/(1.0073) - 1 = 0.0185.$$

11. A is correct. To avoid arbitrage opportunities in exchanging euros and US dollars, investors must be able to lock in a one-year forward exchange rate of USD/

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EUR1.051 today. The solution methodology is shown below.

In one year, a single unit of euro invested risk-free is worth EUR1.0075 ($=e^{0.0075}$).

In one year, a single unit of euro converted to US dollars and then invested risk-free is worth USD1.0589 ($=1.025*e^{0.0325}$).

To convert USD1.0589 into EUR1.0075 requires a forward exchange rate of USD/EUR1.051 (=1.0589/1.0075).

12. A is correct. Regardless of whether the stock increases or decreases in price, the investor's portfolio has a value of USD7.50 as follows:

    If stock price goes to USD35, value = 0.5×35 – 10 = 7.50.

    If stock price goes to USD15, value = 0.5×15 – 0 = 7.50.

    If the stock price rises to USD35, the sold call option at USD25 has a value to the buyer of USD10, offsetting the rise in the stock price.

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PRACTICE PROBLEMS

The following information relates to questions 1-5

Consider the results of an analysis focusing on the market capitalizations of a sample of 100 firms:

Exhibit 1: Market Capitalization of a Sample of 100 Firms

Bin	Cumulative Percentage of Sample (%)	Market Capitalization (euro billions)		Number of Observations
		Lower Bound	Upper Bound	
1	5	0.28	15.45	5
2	10	15.45	21.22	5
3	15	21.22	29.37	5
4	20	29.37	32.57	5
5	25	32.57	34.72	5
6	30	34.72	37.58	5
7	35	37.58	39.90	5
8	40	39.90	41.57	5
9	45	41.57	44.86	5
10	50	44.86	46.88	5
11	55	46.88	49.40	5
12	60	49.40	51.27	5
13	65	51.27	53.58	5
14	70	53.58	56.66	5
15	75	56.66	58.34	5
16	80	58.34	63.10	5
17	85	63.10	67.06	5
18	90	67.06	73.00	5
19	95	73.00	81.62	5
20	100	81.62	96.85	5

1. The tenth percentile corresponds to observations in bin(s):

   A. 2.

   B. 1 and 2.

   C. 19 and 20.

2. The second quintile corresponds to observations in bin(s):

   A. 8.

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B. 5, 6, 7, and 8.

C. 6, 7, 8, 9, and 10.

3. The fourth quartile corresponds to observations in bin(s):

A. 17.

B. 17, 18, 19, and 20.

C. 16, 17, 18, 19, and 20.

4. The median is *closest* to:

A. 44.86.

B. 46.88.

C. 49.40.

5. The interquartile range is *closest* to:

A. 20.76.

B. 23.62.

C. 25.52.

---

6. Exhibit 1 shows the annual MSCI World Index total returns for a 10-year period.

Exhibit 1: MSCI World Index Returns

Year 1	15.25%	Year 6	30.79%
Year 2	10.02%	Year 7	12.34%
Year 3	20.65%	Year 8	–5.02%
Year 4	9.57%	Year 9	16.54%
Year 5	–40.33%	Year 10	27.37%

The fourth quintile return for the MSCI World Index is *closest* to:

A. 20.65 percent.

B. 26.03 percent.

C. 27.37 percent.

The following information relates to questions 7-9

A fund had the following experience over the past 10 years:

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Exhibit 1: Performance over 10 Years

Year	Return
1	4.5%
2	6.0%
3	1.5%
4	–2.0%
5	0.0%
6	4.5%
7	3.5%
8	2.5%
9	5.5%
10	4.0%

7. The fund's standard deviation of returns over the 10 years is *closest* to:

   A. 2.40 percent.

   B. 2.53 percent.

   C. 7.58 percent.

8. The target semideviation of the returns over the 10 years, if the target is 2 percent, is *closest* to:

   A. 1.42 percent.

   B. 1.50 percent.

   C. 2.01 percent.

9. Consider the mean monthly return and the standard deviation for three industry sectors, as shown in Exhibit 2:

Exhibit 2: Mean Monthly Return and Standard Deviations

Sector	Mean Monthly Return (%)	Standard Deviation of Return (%)
Utilities (UTIL)	2.10	1.23
Materials (MATR)	1.25	1.35
Industrials (INDU)	3.01	1.52

Based on the coefficient of variation (CV), the riskiest sector is:

A. utilities.

B. materials.

C. industrials.

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SOLUTIONS

1. B is correct. The tenth percentile corresponds to the lowest 10 percent of the observations in the sample, which are in bins 1 and 2.

2. B is correct. The second quintile corresponds to the second 20 percent of observations. The first 20 percent consists of bins 1 through 4. The second 20 percent of observations consists of bins 5 through 8.

3. C is correct. A quartile consists of 25 percent of the data, and the last 25 percent of the 20 bins are 16 through 20.

4. B is correct. The center of the 20 bins is represented by the market capitalization of the highest value of the 10th bin and the lowest value of the 11th bin, which is 46.88.

5. B is correct. The interquartile range is the difference between the lowest value in the second quartile and the highest value in the third quartile. The lowest value of the second quartile is 34.72, and the highest value of the third quartile is 58.34. Therefore, the interquartile range is 58.34 – 34.72 = 23.62.

6. B is correct. Quintiles divide a distribution into fifths, with the fourth quintile occurring at the point at which 80 percent of the observations lie below it. The fourth quintile is equivalent to the 80th percentile. To find the yth percentile (P_y), we first must determine its location. The formula for the location (L_y) of a yth percentile in an array with n entries sorted in ascending order is $L_y = (n + 1) \times (y/100)$. In this case, $n = 10$ and $y = 80\%$, so

   $L_{80} = (10 + 1) \times (80/100) = 11 \times 0.8 = 8.8.$

   With the data arranged in ascending order (–40.33 percent, –5.02 percent, 9.57 percent, 10.02 percent, 12.34 percent, 15.25 percent, 16.54 percent, 20.65 percent, 27.37 percent, and 30.79 percent), the 8.8th position would be between the eighth and ninth entries, 20.65 percent and 27.37 percent, respectively. Using linear interpolation, $P_{80} = X_8 + (L_y - 8) \times (X_9 - X_8)$,

   $P_{80} = 20.65 + (8.8 - 8) \times (27.37 - 20.65)$

   $= 20.65 + (0.8 \times 6.72) = 20.65 + 5.38$

   $= 26.03$ percent.

7. B is correct. The fund's standard deviation of returns is calculated as follows:

Year	Return	Deviation from Mean	Deviation Squared
1	4.5%	0.0150	0.000225
2	6.0%	0.0300	0.000900
3	1.5%	–0.0150	0.000225
4	–2.0%	–0.0500	0.002500
5	0.0%	–0.0300	0.000900
6	4.5%	0.0150	0.000225
7	3.5%	0.0050	0.000025
8	2.5%	–0.0050	0.000025
9	5.5%	0.0250	0.000625

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Year	Return	Deviation from Mean	Deviation Squared
10	4.0%	0.0100	0.000100
Mean	3.0%		
Sum			**0.005750**

The standard deviation is the square root of the sum of the squared deviations, divided by $n - 1$:

$$s = \sqrt{\frac{0.005750}{(10-1)}} = 2.5276\%.$$

8. B is correct. The target semideviation of the returns over the 10 years with a target of 2 percent is calculated as follows:

Year	Return	Deviation Squared below Target of 2%
1	4.5%	
2	6.0%	
3	1.5%	0.000025
4	–2.0%	0.001600
5	0.0%	0.000400
6	4.5%	
7	3.5%	
8	2.5%	
9	5.5%	
10	4.0%	
Sum		0.002025

The target semideviation is the square root of the sum of the squared deviations from the target, divided by $n - 1$:

$$s_{\text{Target}} = \sqrt{\frac{0.002025}{(10-1)}} = 1.5\%.$$

9. B is correct. The CV is the ratio of the standard deviation to the mean, where a higher CV implies greater risk per unit of return.

$$\text{CV}_{UTIL} = \frac{s}{\bar{X}} = \frac{1.23\%}{2.10\%} = 0.59,$$

$$\text{CV}_{MATR} = \frac{s}{\bar{X}} = \frac{1.35\%}{1.25\%} = 1.08,$$

$$\text{CV}_{INDU} = \frac{s}{\bar{X}} = \frac{1.52\%}{3.01\%} = 0.51.$$

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PRACTICE PROBLEMS

1. An analyst developed two scenarios with respect to the recovery of USD100,000 principal from defaulted loans:

Scenario	Probability of Scenario (%)	Amount Recovered (USD)	Probability of Amount (%)
1	40	50,000	60
		30,000	40
2	60	80,000	90
		60,000	10

The amount of the expected recovery is *closest* to which of the following?

A. USD36,400.

B. USD55,000.

C. USD63,600.

2. The probability distribution for a company's sales is:

Probability	Sales (USD, millions)
0.05	70
0.70	40
0.25	25

The standard deviation of sales is *closest* to which of the following?

A. USD9.81 million.

B. USD12.20 million.

C. USD32.40 million.

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SOLUTIONS

1. C is correct. If Scenario 1 occurs, the expected recovery is 60% (USD50,000) + 40% (USD30,000) = USD42,000, and if Scenario 2 occurs, the expected recovery is 90% (USD80,000) + 10% (USD60,000) = USD78,000. Weighting by the probability of each scenario, the expected recovery is 40% (USD42,000) + 60% (USD78,000) = USD63,600. Alternatively, first calculating the probability of each amount occurring, the expected recovery is (40%)(60%)(USD50,000) + (40%)(40%)(USD30,000) + (60%)(90%)(USD80,000) + (60%)(10%)(USD60,000) = USD63,600.

2. A is correct. The analyst must first calculate expected sales as 0.05 × USD70 + 0.70 × USD40 + 0.25 × USD25 = USD3.50 million + USD28.00 million + USD6.25 million = USD37.75 million.

   After calculating expected sales, we can calculate the variance of sales:

   σ^2 (Sales)

   = P(USD70)[USD70 – E(Sales)]^2 + P(USD40)[USD40 – E(Sales)]^2 + P(USD25)[USD25 – E(Sales)]^2

   = 0.05(USD70 – 37.75)^2 + 0.70(USD40 – 37.75)^2 + 0.25(USD25 – 37.75)^2

   = USD52.00 million + USD3.54 million + USD40.64 million

   = USD96.18 million.

   The standard deviation of sales is thus σ = (USD96.18)$^{1/2}$ = USD9.81 million.

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PRACTICE PROBLEMS

1. An analyst produces the following joint probability function for a foreign index (FI) and a domestic index (DI).

	$R_{DI} = 30\%$	$R_{DI} = 25\%$	$R_{DI} = 15\%$
$R_{FI} = 25\%$	0.25		
$R_{FI} = 15\%$		0.50	
$R_{FI} = 10\%$			0.25

The covariance of returns on the foreign index and the returns on the domestic index is *closest* to:

A. 26.39.

B. 26.56.

C. 28.12.

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SOLUTIONS

1. B is correct. The covariance is 26.56, calculated as follows. First, expected returns are

   $E(R_{FI}) = (0.25 \times 25) + (0.50 \times 15) + (0.25 \times 10)$

   $= 6.25 + 7.50 + 2.50 = 16.25$ and

   $E(R_{DI}) = (0.25 \times 30) + (0.50 \times 25) + (0.25 \times 15)$

   $= 7.50 + 12.50 + 3.75 = 23.75.$

   Covariance is

   $$\text{Cov}(R_{FI},R_{DI}) = \sum_i \sum_j P\left(R_{FI,i},R_{DI,j}\right)\left(R_{FI,i} - E\,R_{FI}\right)\left(R_{DI,j} - E\,R_{DI}\right)$$

   $= 0.25[(25 - 16.25)(30 - 23.75)] + 0.50[(15 - 16.25)(25 - 23.75)] + 0.25[(10 - 16.25)(15 - 23.75)]$

   $= 13.67 + (-0.78) + 13.67 = 26.56.$

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PRACTICE PROBLEMS

1. The weekly closing prices of Mordice Corporation shares are as follows:

Exhibit 1: Mordice Corporation Shares

Date	Closing Price (euros)
1 August	112
8 August	160
15 August	120

The continuously compounded return of Mordice Corporation shares for the period August 1 to August 15 is *closest* to:

A. 6.90 percent.

B. 7.14 percent.

C. 8.95 percent.

2. In contrast to normal distributions, lognormal distributions:

A. are skewed to the left.

B. have outcomes that cannot be negative.

C. are more suitable for describing asset returns than asset prices.

3. The lognormal distribution is a more accurate model for the distribution of stock prices than the normal distribution because stock prices are:

A. symmetrical.

B. unbounded.

C. non-negative.

4. Analysts performing bootstrap:

A. seek to create statistical inferences of population parameters from a single sample.

B. repeatedly draw samples of the same size, with replacement, from the original population.

C. must specify probability distributions for key risk factors that drive the underlying random variables.

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SOLUTIONS

1. A is correct. The continuously compounded return of an asset over a period is equal to the natural log of the asset's price change during the period. In this case, ln(120/112) = 6.90%.

   Note that the continuously compounded return from period 0 to period T is the sum of the incremental one-period continuously compounded returns, which in this case are weekly returns. Specifically:

   Week 1 return: ln(160/112) = 35.67%.

   Week 2 return: ln(120/160) = –28.77%.

   Continuously compounded return = 35.67% + –28.77% = 6.90%.

2. B is correct. By definition, lognormal random variables cannot have negative values (bounded below by 0) and have distributions that are skewed to the right.

3. C is correct. A lognormal distributed variable has a lower bound of zero. The lognormal distribution is also right skewed, which is a useful property in describing asset prices.

4. A is correct. Bootstrapping through random sampling generates the observed variable from a random sampling with unknown population parameters. The analyst does not know the true population distribution, but through sampling can infer the population parameters from the randomly generated sample. B is incorrect because, when performing bootstrap, the analyst repeatedly draws samples from the original sample and not population, where each individual resample has the same size as the original sample and each item drawn is replaced for the next draw. C is incorrect because, when performing bootstrap, analysts simply use the observed empirical distribution of the underlying variables. In a Monte Carlo simulation, in contrast, the analyst would specify probability distributions for key risk factors that drive the underlying variables.

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PRACTICE PROBLEMS

1. Which one of the following statements is true about non-probability sampling?
   A. There is significant risk that the sample is not representative of the population.
   B. Every member of the population has an equal chance of being selected for the sample.
   C. Using judgment guarantees that population subdivisions of interest are represented in the sample.

2. The *best* approach for creating a stratified random sample of a population involves:
   A. drawing an equal number of simple random samples from each subpopulation.
   B. selecting every *k*th member of the population until the desired sample size is reached.
   C. drawing simple random samples from each subpopulation in sizes proportional to the relative size of each subpopulation.

3. A population has a non-normal distribution with mean μ and variance σ^2. The sampling distribution of the sample mean computed from samples of large size from that population will have:
   A. the same distribution as the population distribution.
   B. its mean approximately equal to the population mean.
   C. its variance approximately equal to the population variance.

4. A sample mean is computed from a population with a variance of 2.45. The sample size is 40. The standard error of the sample mean is *closest* to:
   A. 0.039.
   B. 0.247.
   C. 0.387.

5. Compared with bootstrap resampling, jackknife resampling:
   A. is done with replacement.
   B. usually requires that the number of repetitions is equal to the sample size.
   C. produces dissimilar results for every run because resamples are randomly drawn.

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SOLUTIONS

1. A is correct. Because non-probability sampling is dependent on factors other than probability considerations, such as a sampler's judgment or the convenience to access data, there is a significant risk that non-probability sampling might generate a non-representative sample.

2. C is correct. Stratified random sampling involves dividing a population into subpopulations based on one or more classification criteria. Then, simple random samples are drawn from each subpopulation in sizes proportional to the relative size of each subpopulation. These samples are then pooled to form a stratified random sample.

3. B is correct. Given a population described by any probability distribution (normal or non-normal) with finite variance, the central limit theorem states that the sampling distribution of the sample mean will be approximately normal, with the mean approximately equal to the population mean, when the sample size is large.

4. B is correct. Taking the square root of the known population variance to determine the population standard deviation (σ) results in

   $\sigma = \sqrt{2.45} = 1.565.$

   The formula for the standard error of the sample mean (σ_X), based on a known sample size (n), is

   $\sigma_X = \frac{\sigma}{\sqrt{n}}.$

   Therefore,

   $\sigma_X = \frac{1.565}{\sqrt{40}} = 0.247.$

5. B is correct. For a sample of size n, jackknife resampling usually requires n repetitions. In contrast, with bootstrap resampling, we are left to determine how many repetitions are appropriate.

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PRACTICE PROBLEMS

1. An analyst suspects that, in the most recent year, excess returns on stocks have fallen below 5%. She wants to study whether the excess returns are less than 5%. Designating the population mean as μ, which hypotheses are most appropriate for her analysis?

   A. H_0: $\mu = 5\%$ versus H_a: $\mu \neq 5\%$

   B. H_0: $\mu \geq 5\%$ versus H_a: $\mu < 5\%$

   C. H_0: $\mu \leq 5\%$ versus H_a: $\mu > 5\%$

2. Which of the following statements about hypothesis testing is correct?

   A. The null hypothesis is the condition a researcher hopes to support.

   B. The alternative hypothesis is the proposition considered true without conclusive evidence to the contrary.

   C. The alternative hypothesis exhausts all potential parameter values not accounted for by the null hypothesis.

3. Which of the following statements regarding the null hypothesis is correct?

   A. It can be stated as "not equal to" provided the alternative hypothesis is stated as "equal to."

   B. Along with the alternative hypothesis, it considers all possible values of the population parameter.

   C. In a two-tailed test, it is rejected when evidence supports equality between the hypothesized value and the population parameter.

4. Which of the following statements regarding a one-tailed hypothesis test is correct?

   A. The rejection region increases in size as the level of significance becomes smaller.

   B. A one-tailed test more strongly reflects the beliefs of the researcher than a two-tailed test.

   C. The absolute value of the critical value is larger than that for a two-tailed test at the same level of significance.

5. If a researcher selects a 5 percent level of significance for a hypothesis test, the confidence level is:

   A. 2.5 percent.

   B. 5 percent.

   C. 95 percent.

6. A hypothesis test for a normally distributed population, at a 0.05 significance

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level, implies a:

A. 95 percent probability of rejecting a true null hypothesis.

B. 95 percent probability of a Type I error for a two-tailed test.

C. 5 percent critical value rejection region for a one-tailed test.

7. The value of a test statistic is *best* described as the basis for deciding whether to:

A. reject the null hypothesis.

B. accept the null hypothesis.

C. reject the alternative hypothesis.

8. Which of the following *best* describes a Type I error?

A. Rejecting a true null hypothesis

B. Rejecting a false null hypothesis

C. Failing to reject a false null hypothesis

9. A Type II error is *best* described as:

A. rejecting a true null hypothesis.

B. failing to reject a false null hypothesis.

C. failing to reject a false alternative hypothesis.

10. The level of significance of a hypothesis test is *best* used to:

A. calculate the test statistic.

B. define the test's rejection points.

C. specify the probability of a Type II error.

11. The probability of correctly rejecting the null hypothesis is the:

A. *p*-value.

B. power of a test.

C. level of significance.

12. The power of a hypothesis test is:

A. equivalent to the level of significance.

B. the probability of not making a Type II error.

C. unchanged by increasing a small sample size.

13. In the step "stating a decision rule" in testing a hypothesis, which of the following elements must be specified?

A. Critical value

B. Power of a test

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C. Value of a test statistic

14. A pooled estimator is used when testing a hypothesis concerning the:

A. equality of the variances of two normally distributed populations.

B. difference between the means of two approximately normally distributed populations with unknown but assumed equal variances.

C. difference between the means of two at least approximately normally distributed populations with unknown and assumed unequal variances.

15. When evaluating mean differences between two dependent samples, the *most* appropriate test is a:

A. *z*-test.

B. chi-square test.

C. paired comparisons test.

16. A chi-square test is *most* appropriate for tests concerning:

A. a single variance.

B. differences between two population means with variances assumed to be equal.

C. differences between two population means with variances not assumed to be equal.

17. Which of the following tests should be used to test the difference between the variances of two normally distributed populations with random independent samples?

A. *t*-test

B. *F*-test

C. Paired comparisons test

18. A nonparametric test is most appropriate when the:

A. data consist of ranked values.

B. validity of the test depends on many assumptions.

C. sample sizes are large but are drawn from a population that may be non-normal.

19. In which of the following situations would a nonparametric test of a hypothesis *most likely* be used?

A. The sample data are ranked according to magnitude.

B. The sample data come from a normally distributed population.

C. The test validity depends on many assumptions about the nature of the population.

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20. An analyst is examining the monthly returns for two funds over one year. Both funds' returns are non-normally distributed. To test whether the mean return of one fund is greater than the mean return of the other fund, the analyst can use:

A. a parametric test only.

B. a nonparametric test only.

C. both parametric and nonparametric tests.

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SOLUTIONS

1. B is correct. The null hypothesis is what she wants to reject in favor of the alternative, which is that population mean excess return is less than 5%. This is a one-sided (left-side) alternative hypothesis.

2. C is correct. Together, the null and alternative hypotheses account for all possible values of the parameter. Any possible values of the parameter not covered by the null must be covered by the alternative hypothesis (e.g., H_0: $\mu \leq 5$ versus H_a: $\mu > 5$). A is incorrect because the null hypothesis is what the researcher wants to reject; the "hoped for" or "suspected" condition is often set up as the alternative hypothesis. B is incorrect because the null (not the alternative) hypothesis is considered to be true unless the sample used to conduct the hypothesis test gives convincing evidence that the null hypothesis is false.

3. B is correct. The null and alternative hypotheses are complements of one another and must be both mutually and collectively exhaustive. Differently put: all possible values or outcomes need to be contained in either the null or the alternative hypothesis.

   A is incorrect because the null hypothesis must always include the equality sign (less than or equal to, equal to, or greater than or equal to). C is incorrect because, in a two-tailed test, the null hypothesis is generally set up as equality between the hypothesized value and the population parameter. If evidence supports equality, then the null hypothesis would not be rejected.

4. B is correct. One-tailed tests in which the alternative is "greater than" or "less than" represent the beliefs of the researcher more firmly than a "not equal to" alternative hypothesis.

   A is incorrect because a smaller significance level implies a smaller rejection region. C is incorrect because the absolute value of the critical value for a one-tailed hypothesis test is smaller than that of a two-tailed test.

   For example, for a two-tailed t-test with 30 degrees of freedom at the 5% significance level, the corresponding critical value is ±2.042 (2.5% in each tail) whereas the corresponding critical value for a one-tailed t-test is +1.697 or –1.697 (5% in the left or right tail). Thus, the absolute value of the critical value is smaller for the one-tailed test than it is for the two-tailed test for the same level of significance.

5. C is correct. The 5 percent level of significance (i.e., probability of a Type I error) corresponds to 1 – 0.05 = 0.95, or a 95 percent confidence level (i.e., probability of not rejecting a true null hypothesis). The level of significance is the complement to the confidence level; in other words, they sum to 1.00, or 100 percent.

6. C is correct. For a one-tailed hypothesis test, there is a 5 percent rejection region in one tail of the distribution. A is incorrect because a 5 percent significance level implies a 5 percent probability of rejecting the null hypothesis and a 95 percent confidence interval. B is incorrect because the probability of a Type I error (mistakenly rejecting a true null) is the stated 5 percent significance level.

7. A is correct. In hypothesis testing, a test statistic is a quantity whose value is the basis for deciding whether to reject the null hypothesis.

8. A is correct. The definition of a Type I error is when a true null hypothesis is rejected.

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9. B is correct. A Type II error occurs when a false null hypothesis is not rejected.

10. B is correct. The level of significance is used to establish the rejection points of the hypothesis test. A is correct because the significance level is not used to calculate the test statistic; rather, it is used to determine the critical value. C is incorrect because the significance level specifies the probability of making a Type I error.

11. B is correct. The power of a test is the probability of rejecting the null hypothesis when it is false. A is incorrect because the p-value is the smallest level of significance at which the null hypothesis can be rejected. C is incorrect because the level of significance is the probability of mistakenly rejecting the null hypothesis (Type I error).

12. B is correct. The power of a hypothesis test is the probability of correctly rejecting the null when it is false. Failing to reject the null when it is false is a Type II error. Thus, the power of a hypothesis test is the probability of not committing a Type II error.

13. A is correct. The critical value in a decision rule is the rejection point for the test. It is the point with which the test statistic is compared to determine whether to reject the null hypothesis, which is part of the fourth step in hypothesis testing. B is incorrect because the power of a test refers to the probability of rejecting the null hypothesis when it is false. C is incorrect because the value of the test statistic is specified in the 'Identify the appropriate test statistic and its probability distribution" step.

14. B is correct. The assumption that the variances are equal allows for the combining of both samples to obtain a pooled estimate of the common variance.

15. C is correct. A paired comparisons test is appropriate to test the mean differences of two samples believed to be dependent. A is incorrect because a z-test is used to determine whether two population means are different when the variances are known and the sample size is large. B is incorrect because a chi-square test is used for tests concerning the variance of a single normally distributed population.

16. A is correct. A chi-square test is used for tests concerning the variance of a single normally distributed population.

17. B is correct. An *F*-test is used to conduct tests concerning the difference between the variances of two normally distributed populations with random independent samples.

18. A is correct. When the samples consist of ranked values, parametric tests are not appropriate. In such cases, nonparametric tests are most appropriate.

19. A is correct. A nonparametric test is used when the data are given in ranks.

20. B is correct. There are only 12 (monthly) observations over the one year of the sample and thus the samples are small. Additionally, the funds' returns are non-normally distributed. Therefore, the samples do not meet the distributional assumptions for a parametric test. The Mann–Whitney U test (a nonparametric test) could be used to test the differences between population means.

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PRACTICE PROBLEMS

1. Jill Batten is analyzing how the returns on the stock of Stellar Energy Corp. are related with the previous month's percentage change in the US Consumer Price Index for Energy (CPIENG). Based on 248 observations, she has computed the sample correlation between the Stellar and CPIENG variables to be –0.1452. She also wants to determine whether the sample correlation is significantly different from zero. The critical value for the test statistic at the 0.05 level of significance is approximately 1.96. Batten should conclude that the statistical relationship between Stellar and CPIENG is:

   A. significant, because the calculated test statistic is outside the bounds of the critical values for the test statistic.

   B. significant, because the calculated test statistic has a lower absolute value than the critical value for the test statistic.

   C. insignificant, because the calculated test statistic is outside the bounds of the critical values for the test statistic.

2. Which of the following statements is correct regarding the chi-square test of independence?

   A. The test has a one-sided rejection region.

   B. The null hypothesis is that the two groups are dependent.

   C. If there are two categories, each with three levels or groups, there are six degrees of freedom.

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SOLUTIONS

1. A is correct. The calculated test statistic is

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$
$$= \frac{-0.1452\sqrt{248-2}}{\sqrt{1-(-0.1452)^2}} = -2.30177.$$

Because the value of $t = -2.30177$ is outside the bounds of ±1.96, we reject the null hypothesis of no correlation and conclude that evidence is sufficient to indicate that the correlation is different from zero.

2. A is correct. The test statistic includes squared differences between the observed and expected values, so the test involves only one side, the right side. B is incorrect because the null hypothesis is that the groups are independent, and C is incorrect because with three levels of groups for the two categorical variables, there are four degrees of freedom.

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PRACTICE PROBLEMS

The following information relates to questions 1-3

An analyst has estimated a model that regresses a company's return on equity (ROE) against its growth opportunities (GO), defined as the company's three-year compounded annual growth rate in sales, over 20 years, and produces the following estimated simple linear regression:

$ROE_i = 4 + 1.8\ GO_i + \varepsilon_i$.

Both variables are stated in percentages, so a GO observation of 5 percent is included as 5.

1. The predicted value of the company's ROE if its GO is 10 percent is closest to:
   - **A.** 1.8 percent.
   - **B.** 15.8 percent.
   - **C.** 22.0 percent.

2. The change in ROE for a change in GO from 5 percent to 6 percent is closest to:
   - **A.** 1.8 percent.
   - **B.** 4.0 percent.
   - **C.** 5.8 percent.

3. The residual in the case of a *GO* of 8 percent and an observed *ROE* of 21 percent is closest to:
   - **A.** −1.8 percent.
   - **B.** 2.6 percent.
   - **C.** 12.0 percent.

---

4. Homoskedasticity is best described as the situation in which the variance of the residuals of a regression is:
   - **A.** zero.
   - **B.** normally distributed.
   - **C.** constant across observations.

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The following information relates to questions 5-8

An analyst is examining the annual growth of the money supply for a country over the past 30 years. This country experienced a central bank policy shift 15 years ago, which altered the approach to the management of the money supply. The analyst estimated a model using the annual growth rate in the money supply regressed on the variable (SHIFT) that takes on a value of 0 before the policy shift and 1 after. She estimated the values in Exhibit 1:

Exhibit 1: SHIFT Estimates

	Coefficients	Standard Error	*t*-Stat.
Intercept	5.767264	0.445229	12.95348
SHIFT	–5.13912	0.629649	–8.16188

Critical t-values, level of significance of 0.05:
eOne-sided, left side: –1.701
fOne-sided, right side: +1.701
gTwo-sided: ±2.048

5. The variable SHIFT is best described as:
   A. an indicator variable.
   B. a dependent variable.
   C. a continuous variable.

6. The interpretation of the intercept is the mean of the annual growth rate of the money supply:
   A. before the shift in policy.
   B. over the entire period.
   C. after the shift in policy.

7. The interpretation of the slope is the:
   A. change in the annual growth rate of the money supply per year.
   B. average annual growth rate of the money supply after the shift in policy.
   C. difference in the average annual growth rate of the money supply from before to after the shift in policy.

8. Testing whether there is a change in the money supply growth after the shift in policy, using a 0.05 level of significance, we conclude that there is:
   A. sufficient evidence that the money supply growth changed.
   B. not enough evidence that the money supply growth is different from zero.
   C. not enough evidence to indicate that the money supply growth changed.

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The following information relates to questions 9-12

Kenneth McCoin, CFA, is a challenging interviewer. Last year, he handed each job applicant a sheet of paper with the information in Exhibit 1, and he then asked several questions about regression analysis. Some of McCoin's questions, along with a sample of the answers he received to each, are given below. McCoin told the applicants that the independent variable is the ratio of net income to sales for restaurants with a market cap of more than $100 million and the dependent variable is the ratio of cash flow from operations to sales for those restaurants. Which of the choices provided is the best answer to each of McCoin's questions?

Exhibit 1: Regression Analysis

Regression Statistics	
R^2	0.7436
Standard error	0.0213
Observations	24

Source	df	Sum of Squares	Mean Square	*F*	*p*-Value
Regression	1	0.029	0.029000	63.81	0
Residual	22	0.010	0.000455		
Total	23	0.040			

	Coefficients	Standard Error	*t*-Statistic	*p*-Value
Intercept	0.077	0.007	11.328	0
Net income to sales (%)	0.826	0.103	7.988	0

9. The coefficient of determination is *closest* to:

A. 0.7436.

B. 0.8261.

C. 0.8623.

10. The correlation between X and Y is *closest* to:

A. −0.7436.

B. 0.7436.

C. 0.8623.

11. If the ratio of net income to sales for a restaurant is 5 percent, the predicted ratio of cash flow from operations (CFO) to sales is *closest* to:

A. −4.054.

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B. 0.524.

C. 4.207.

12. Is the relationship between the ratio of cash flow to operations and the ratio of net income to sales significant at the 0.05 level?

A. No, because the R^2 is greater than 0.05

B. No, because the *p*-values of the intercept and slope are less than 0.05

C. Yes, because the *p*-values for *F* and *t* for the slope coefficient are less than 0.05

The following information relates to questions 13-17

Howard Golub, CFA, is preparing to write a research report on Stellar Energy Corp. common stock. One of the world's largest companies, Stellar is in the business of refining and marketing oil. As part of his analysis, Golub wants to evaluate the sensitivity of the stock's returns to various economic factors. For example, a client recently asked Golub whether the price of Stellar Energy Corp. stock has tended to rise following increases in retail energy prices. Golub believes the association between the two variables is negative, but he does not know the strength of the association.

Golub directs his assistant, Jill Batten, to study the relationships between (1) Stellar monthly common stock returns and the previous month's percentage change in the US Consumer Price Index for Energy (CPIENG) and (2) Stellar monthly common stock returns and the previous month's percentage change in the US Producer Price Index for Crude Energy Materials (PPICEM). Golub wants Batten to run both a correlation and a linear regression analysis. In response, Batten compiles the summary statistics shown in Exhibit 1 for 248 months. All the data are in decimal form, where 0.01 indicates a 1 percent return. Batten also runs a regression analysis using Stellar monthly returns as the dependent variable and the monthly change in CPIENG as the independent variable. Exhibit 2 displays the results of this regression model.

Exhibit 1: Descriptive Statistics

	Stellar Common Stock Monthly Return	Lagged Monthly Change	
		CPIENG	PPICEM
Mean	0.0123	0.0023	0.0042
Standard deviation	0.0717	0.0160	0.0534
Covariance, Stellar vs. CPIENG	−0.00017		
Covariance, Stellar vs. PPICEM	−0.00048		
Covariance, CPIENG vs. PPICEM	0.00044		
Correlation, Stellar vs. CPIENG	−0.1452		

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Exhibit 2: Regression Analysis with CPIENG

Regression Statistics	
R^2	0.0211
Standard error of the estimate	0.0710
Observations	248

	Coefficients	Standard Error	*t*-Statistic
Intercept	0.0138	0.0046	3.0275
CPIENG (%)	−0.6486	0.2818	−2.3014

Critical t-values
One-sided, left side: −1.651
One-sided, right side: +1.651
Two-sided: ±1.967

13. Which of the following best describes Batten's regression?

A. Time-series regression

B. Cross-sectional regression

C. Time-series and cross-sectional regression

14. Based on the regression, if the CPIENG *decreases* by 1.0 percent, the expected return on Stellar common stock during the next period is *closest* to:

A. 0.0073 (0.73 percent).

B. 0.0138 (1.38 percent).

C. 0.0203 (2.03 percent).

15. Based on Batten's regression model, the coefficient of determination indicates that:

A. Stellar's returns explain 2.11 percent of the variability in CPIENG.

B. Stellar's returns explain 14.52 percent of the variability in CPIENG.

C. changes in CPIENG explain 2.11 percent of the variability in Stellar's returns.

16. For Batten's regression model, 0.0710 is the standard deviation of:

A. the dependent variable.

B. the residuals from the regression.

C. the predicted dependent variable from the regression.

17. For the analysis run by Batten, which of the following is an *incorrect* conclusion from the regression output?

A. The estimated intercept from Batten's regression is statistically different from zero at the 0.05 level of significance.

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B. In the month after the CPIENG declines, Stellar's common stock is expected to exhibit a positive return.

C. Viewed in combination, the slope and intercept coefficients from Batten's regression are not statistically different from zero at the 0.05 level of significance.

The following information relates to questions 18-26

Anh Liu is an analyst researching whether a company's debt burden affects investors' decision to short the company's stock. She calculates the short interest ratio (the ratio of short interest to average daily share volume, expressed in days) for 50 companies as of the end of the year and compares this ratio with the companies' debt ratio (the ratio of total liabilities to total assets, expressed in decimal form).

Liu provides a number of statistics in Exhibit 1. She also estimates a simple regression to investigate the effect of the debt ratio on a company's short interest ratio. The results of this simple regression, including the analysis of variance (ANOVA), are shown in Exhibit 2.

In addition to estimating a regression equation, Liu graphs the 50 observations using a scatter plot, with the short interest ratio on the vertical axis and the debt ratio on the horizontal axis.

Exhibit 1: Summary Statistics

Statistic	Debt Ratio (X_i)	Short Interest Ratio (Y_i)
Sum	19.8550	192.3000
Sum of squared deviations from the mean	$\sum_{i=1}^{n}(X_i - \overline{X})^2 = 2.2225.$	$\sum_{i=1}^{n}(Y_i - \overline{Y})^2 = 412.2042.$
Sum of cross-products of deviations from the mean	$\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y}) = -9.2430.$	

Exhibit 2: Regression of the Short Interest Ratio on the Debt Ratio

ANOVA	Degrees of Freedom (df)	Sum of Squares	Mean Square
Regression	1	38.4404	38.4404
Residual	48	373.7638	7.7867
Total	49	412.2042	
Regression Statistics			
R^2	0.0933		
Standard error of estimate	2.7905		

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ANOVA	Degrees of Freedom (df)	Sum of Squares	Mean Square
Observations	50		

	Coefficients	Standard Error	*t*-Statistic
Intercept	5.4975	0.8416	6.5322
Debt ratio (%)	−4.1589	1.8718	−2.2219

Critical t-values for a 0.05 level of significance:
One-sided, left side: −1.677
One-sided, right side: +1.677
Two-sided: ±2.011

Liu is considering three interpretations of these results for her report on the relationship between debt ratios and short interest ratios:

Interpretation 1 Companies' higher debt ratios cause lower short interest ratios.

Interpretation 2 Companies' higher short interest ratios cause higher debt ratios.

Interpretation 3 Companies with higher debt ratios tend to have lower short interest ratios.

She is especially interested in using her estimation results to predict the short interest ratio for MQD Corporation, which has a debt ratio of 0.40.

18. Based on Exhibit 1 and Exhibit 2, if Liu were to graph the 50 observations, the scatter plot summarizing this relation would be *best* described as:

A. horizontal.

B. upward sloping.

C. downward sloping.

19. Based on Exhibit 1, the sample covariance is *closest to*:

A. −9.2430.

B. −0.1886.

C. 8.4123.

20. Based on Exhibit 1 and Exhibit 2, the correlation between the debt ratio and the short interest ratio is *closest to*:

A. −0.3054.

B. 0.0933.

C. 0.3054.

21. Which of the interpretations *best* describes Liu's findings?

A. Interpretation 1

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B. Interpretation 2

C. Interpretation 3

22. The dependent variable in Liu's regression analysis is the:

A. intercept.

B. debt ratio.

C. short interest ratio.

23. Based on Exhibit 2, the number of degrees of freedom for the *t*-test of the slope coefficient in this regression is:

A. 48.

B. 49.

C. 50.

24. Which of the following should Liu conclude from the results shown in Exhibit 2?

A. The average short interest ratio is 5.4975.

B. The estimated slope coefficient is different from zero at the 0.05 level of significance.

C. The debt ratio explains 30.54 percent of the variation in the short interest ratio.

25. Based on Exhibit 2, the short interest ratio expected for MQD Corporation is *closest* to:

A. 3.8339.

B. 5.4975.

C. 6.2462.

26. Based on Liu's regression results in Exhibit 2, the *F*-statistic for testing whether the slope coefficient is equal to zero is *closest* to:

A. –2.2219.

B. 3.5036.

C. 4.9367.

The following information relates to questions 27-29

Doug Abitbol is a portfolio manager for Polyi Investments, a hedge fund that trades in the United States. Abitbol manages the hedge fund with the help of Robert Olabudo, a junior portfolio manager.

Abitbol looks at economists' inflation forecasts and would like to examine the

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relationship between the US Consumer Price Index (US CPI) consensus forecast and the actual US CPI using regression analysis. Olabudo estimates regression coefficients to test whether the consensus forecast is unbiased. If the consensus forecasts are unbiased, the intercept should be 0.0 and the slope will be equal to 1.0. Regression results are presented in Exhibit 1. Additionally, Olabudo calculates the 95 percent prediction interval of the actual CPI using a US CPI consensus forecast of 2.8.

Exhibit 1: Regression Output: Estimating US CPI

Regression Statistics	
R^2	0.9859
Standard error of estimate	0.0009
Observations	60

	Coefficients	**Standard Error**	***t*-Statistic**
Intercept	0.0001	0.0002	0.5000
US CPI consensus forecast	0.9830	0.0155	63.4194

Notes:

1. The absolute value of the critical value for the *t*-statistic is 2.002 at the 5 percent level of significance.
2. The standard deviation of the US CPI consensus forecast is s_x = 0.7539.
3. The mean of the US CPI consensus forecast is $\overline{X}$ = 1.3350.

Finally, Abitbol and Olabudo discuss the forecast and forecast interval:

- Observation 1. For a given confidence level, the forecast interval is the same no matter the US CPI consensus forecast.
- Observation 2. A larger standard error of the estimate will result in a wider confidence interval.

27. Based on Exhibit 1, Olabudo should:

A. conclude that the inflation predictions are unbiased.

B. reject the null hypothesis that the slope coefficient equals one.

C. reject the null hypothesis that the intercept coefficient equals zero.

28. Based on Exhibit 1, Olabudo should calculate a prediction interval for the actual US CPI *closest* to:

A. 2.7506 to 2.7544.

B. 2.7521 to 2.7529.

C. 2.7981 to 2.8019.

29. Which of Olabudo's observations of forecasting is correct?

A. Only Observation 1

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B. Only Observation 2

C. Both Observation 1 and Observations 2

The following information relates to questions 30-34

Elena Vasileva recently joined EnergyInvest as a junior portfolio analyst. Vasileva's supervisor asks her to evaluate a potential investment opportunity in Amtex, a multinational oil and gas corporation based in the United States. Vasileva's supervisor suggests using regression analysis to examine the relation between Amtex shares and returns on crude oil.

Vasileva notes the following assumptions of regression analysis:

- Assumption 1. The error term is uncorrelated across observations.
- Assumption 2. The variance of the error term is the same for all observations.
- Assumption 3. The dependent variable is normally distributed.

Vasileva runs a regression of Amtex share returns on crude oil returns using the monthly data she collected. Selected data used in the regression are presented in Exhibit 1, and selected regression output is presented in Exhibit 2. She uses a 1 percent level of significance in all her tests.

Exhibit 1: Selected Data for Crude Oil Returns and Amtex Share Returns

	Oil Return (X_i)	Amtex Return (Y_i)	Cross-Product $(X_i - \overline{X})(Y_i - \overline{Y})$	Predicted Amtex Return ($\hat{Y}_i$)	Regression Residual ($Y_i - \hat{Y}_i$)	Squared Residual $(Y_i - \hat{Y}_i)^2$
Month 1	–0.032000	0.033145	–0.000388	0.002011	–0.031134	0.000969
⋮	⋮	⋮	⋮	⋮	⋮	⋮
Month 36	0.028636	0.062334	0.002663	0.016282	–0.046053	0.002121
Sum			0.085598			0.071475
Average	–0.018056	0.005293				

Exhibit 2: Selected Regression Output, Dependent Variable: Amtex Share Return

	Coefficient	Standard Error
Intercept	0.0095	0.0078
Oil return	0.2354	0.0760

Critical *t*-values for a 1 percent level of significance:

One-sided, left side: –2.441

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One-sided, right side: +2.441

Two-sided: ±2.728

Vasileva expects the crude oil return next month, Month 37, to be −0.01. She computes the standard error of the forecast to be 0.0469.

30. Which of Vasileva's assumptions regarding regression analysis is *incorrect*?

A. Assumption 1

B. Assumption 2

C. Assumption 3

31. Based on Exhibit 1, the standard error of the estimate is *closest* to:

A. 0.04456.

B. 0.04585.

C. 0.05018.

32. Based on Exhibit 2, Vasileva should reject the null hypothesis that:

A. the slope is less than or equal to 0.15.

B. the intercept is less than or equal to zero.

C. crude oil returns do not explain Amtex share returns.

33. Based on Exhibit 2 and Vasileva's prediction of the crude oil return for Month 37, the estimate of Amtex share return for Month 37 is *closest* to:

A. −0.0024.

B. 0.0071.

C. 0.0119.

34. Using information from Exhibit 2, the 99 percent prediction interval for Amtex share return for Month 37 is *best* described as:

A. $\hat{Y}_f \pm 0.0053$.

B. $\hat{Y}_f \pm 0.0469$.

C. $\hat{Y}_f \pm 0.1279$.

The following information relates to questions 35-38

Espey Jones is examining the relation between the net profit margin (NPM) of companies, in percent, and their fixed asset turnover (FATO). He collected a sample of 35 companies for the most recent fiscal year and fit several different functional forms, settling on the following model:

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$\ln(\text{NPM}_i) = b_0 + b_1 \text{FATO}_i$.

The results of this estimation are provided in Exhibit 1.

Exhibit 1: Results of Regressing NPM on FATO

Source	df	Sum of Squares	Mean Square	*F*	*p*-Value
Regression	1	102.9152	102.9152	1,486.7079	0.0000
Residual	32	2.2152	0.0692		
Total	33	105.1303			

	Coefficients	Standard Error	*t*- Statistic	*p*-Value
Intercept	0.5987	0.0561	10.6749	0.0000
FATO	0.2951	0.0077	38.5579	0.0000

35. The coefficient of determination is *closest* to:

A. 0.0211.

B. 0.9789.

C. 0.9894.

36. The standard error of the estimate is *closest* to:

A. 0.2631.

B. 1.7849.

C. 38.5579.

37. At a 0.01 level of significance, Jones should conclude that:

A. the mean net profit margin is 0.5987 percent.

B. the variation of the fixed asset turnover explains the variation of the natural log of the net profit margin.

C. a change in the fixed asset turnover from three to four times is likely to result in a change in the net profit margin of 0.5987 percent.

38. The predicted net profit margin for a company with a fixed asset turnover of 2 times is *closest* to:

A. 1.1889 percent.

B. 1.8043 percent.

C. 3.2835 percent

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SOLUTIONS

1. C is correct. The predicted value of ROE = 4 + (1.8 × 10) = 22.

2. A is correct. The slope coefficient of 1.8 is the expected change in the dependent variable (*ROE*) for a one-unit change in the independent variable (*GO*).

3. B is correct. The predicted value is *ROE* = 4 + (1.8 × 8) = 18.4. The observed value of *ROE* is 21, so the residual is 2.6 = 21.0 − 18.4.

4. C is correct. Homoskedasticity is the situation in which the variance of the residuals is constant across the observations.

5. A is correct. SHIFT is an indicator or dummy variable because it takes on only the values 0 and 1.

6. A is correct. In a simple regression with a single indicator variable, the intercept is the mean of the dependent variable when the indicator variable takes on a value of zero, which is before the shift in policy in this case.

7. C is correct. Whereas the intercept is the average of the dependent variable when the indicator variable is zero (i.e., before the shift in policy), the slope is the difference in the mean of the dependent variable from before to after the change in policy.

8. A is correct. The null hypothesis of no difference in the annual growth rate is rejected at the 0.05 level: The calculated test statistic of −8.16188 is outside the bounds of ±2.048.

9. A is correct. The coefficient of determination is the same as R^2, which is 0.7436 in the table.

10. C is correct. Because the slope is positive, the correlation between X and Y is simply the square root of the coefficient of determination: $\sqrt{0.7436}$ = 0.8623.

11. C is correct. To make a prediction using the regression model, multiply the slope coefficient by the forecast of the independent variable and add the result to the intercept. Expected value of CFO to sales = 0.077 + (0.826 × 5) = 4.207.

12. C is correct. The *p*-value is the smallest level of significance at which the null hypotheses concerning the slope coefficient can be rejected. In this case, the *p*-value is less than 0.05, and thus the regression of the ratio of cash flow from operations to sales on the ratio of net income to sales is significant at the 5 percent level.

13. A is correct. The data are observations over time.

14. C is correct. From the regression equation, Expected return = 0.0138 + (−0.6486 × −0.01) = 0.0138 + 0.006486 = 0.0203, or 2.03 percent.

15. C is correct. R^2 is the coefficient of determination. In this case, it shows that 2.11% of the variability in Stellar's returns is explained by changes in CPIENG.

16. B is correct. The standard error of the estimate is the standard deviation of the regression residuals.

17. C is the correct response because it is a false statement. The slope and intercept are both statistically different from zero at the 0.05 level of significance.

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18. C is correct. The slope coefficient (shown in Exhibit 2) is negative.

19. B is correct. The sample covariance is calculated as follows:
$$\frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{n-1} = -9.2430 \div 49 = -0.1886.$$

20. A is correct. In simple regression, the R^2 is the square of the pairwise correlation. Because the slope coefficient is negative, the correlation is the negative of the square root of 0.0933, or −0.3055.

21. C is correct. Conclusions cannot be drawn regarding causation; they can be drawn only about association; therefore, Interpretations 1 and 2 are incorrect.

22. C is correct. Liu explains the variation of the short interest ratio using the variation of the debt ratio.

23. A is correct. The degrees of freedom are the number of observations minus the number of parameters estimated, which equals 2 in this case (the intercept and the slope coefficient). The number of degrees of freedom is 50 − 2 = 48.

24. B is correct. The *t*-statistic is −2.2219, which is outside the bounds created by the critical *t*-values of ±2.011 for a two-tailed test with a 5 percent significance level. The value of 2.011 is the critical *t*-value for the 5 percent level of significance (2.5 percent in one tail) for 48 degrees of freedom. A is incorrect because the mean of the short interest ratio is 192.3 ÷ 50 = 3.846. C is incorrect because the debt ratio explains 9.33 percent of the variation of the short interest ratio.

25. A is correct. The predicted value of the short interest ratio = 5.4975 + (−4.1589 × 0.40) = 5.4975 − 1.6636 = 3.8339.

26. C is correct. The calculation is $F = \frac{\text{Mean square regression}}{\text{Mean square error}} = \frac{38.4404}{7.7867} = 4.9367.$

27. A is correct. We fail to reject the null hypothesis of a slope equal to one, and we fail to reject the null hypothesis of an intercept equal to zero. The test of the slope equal to 1.0 is

    $$t = \frac{0.9830 - 1.000}{0.0155} = -1.09677.$$

    The test of the intercept equal to 0.0 is

    $$t = \frac{0.0001 - 0.0000}{.00002} = 0.5000.$$

    Therefore, we conclude that the forecasts are unbiased.

28. A is correct. The forecast interval for inflation is calculated in three steps:

    Step 1. Make the prediction given the US CPI forecast of 2.8:

    $$\widehat{Y} = \widehat{b_0} + \widehat{b_1}X$$
    $$= 0.0001 + (0.9830 \times 2.8)\cdot$$
    $$= 2.7525.$$

    Step 2. Compute the variance of the prediction error:

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$$s_f^2 = s_e^2\left\{1 + (1/n) + \left[\left(X_f - \bar{X}\right)^2\right] / \left[(n-1) \times s_x^2\right]\right\}.$$

$$s_f^2 = 0.0009^2\left\{1 + (1/60) + \left[(2.8 - 1.3350)^2\right] / \left[(60-1) \times 0.7539^2\right]\right\}.$$

$$s_f^2 = 0.00000088.$$

$$s_f = 0.0009.$$

Step 3. Compute the prediction interval:

$\hat{Y} \pm t_c \times s_f$.

$2.7525 \pm (2.0 \times 0.0009)$

Lower bound: $2.7525 - (2.0 \times 0.0009) = 2.7506$.

Upper bound: $2.7525 + (2.0 \times 0.0009) = 2.7544$.

Given the US CPI forecast of 2.8, the 95 percent prediction interval is 2.7506 to 2.7544.

29. B is correct. The confidence level influences the width of the forecast interval through the critical *t*-value that is used to calculate the distance from the forecasted value: The larger the confidence level, the wider the interval. Therefore, Observation 1 is not correct.
Observation 2 is correct. The greater the standard error of the estimate, the greater the standard error of the forecast.

30. C is correct. The assumptions of the linear regression model are that (1) the relationship between the dependent variable and the independent variable is linear in the parameters b_0 and b_1, (2) the residuals are independent of one another, (3) the variance of the error term is the same for all observations, and (4) the error term is normally distributed. Assumption 3 is incorrect because the dependent variable need not be normally distributed.

31. B is correct. The standard error of the estimate for a linear regression model with one independent variable is calculated as the square root of the mean square error:

$$s_e = \sqrt{\frac{0.071475}{34}} = 0.04585.$$

32. C is correct. Crude oil returns explain the Amtex share returns if the slope coefficient is statistically different from zero. The slope coefficient is 0.2354, and the calculated *t*-statistic is

$$t = \frac{0.2354 - 0.0000}{0.0760} = 3.0974,$$

which is outside the bounds of the critical values of ±2.728.
Therefore, Vasileva should reject the null hypothesis that crude oil returns do not explain Amtex share returns, because the slope coefficient is statistically different from zero.
A is incorrect because the calculated *t*-statistic for testing the slope against 0.15 is $t = \frac{0.2354 - 0.1500}{0.0760} = 1.1237$, which is less than the critical value of +2.441.
B is incorrect because the calculated *t*-statistic is $t = \frac{0.0095 - 0.0000}{0.0078} = 1.2179$, which is less than the critical value of +2.441.

33. B is correct. The predicted value of the dependent variable, Amtex share return, given the value of the independent variable, crude oil return, –0.01, is calculated

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as $\hat{Y} = \hat{b}_0 + \hat{b}_1 X_i = 0.0095 + [0.2354 \times (-0.01)] = 0.0071$.

34. C is correct. The predicted share return is 0.0095 + [0.2354 × (–0.01)] = 0.0071. The lower limit for the prediction interval is 0.0071 – (2.728 × 0.0469) = –0.1208, and the upper limit for the prediction interval is 0.0071 + (2.728 × 0.0469) = 0.1350.

    A is incorrect because the bounds of the interval should be based on the standard error of the forecast and the critical *t*-value, not on the mean of the dependent variable.

    B is incorrect because bounds of the interval are based on the product of the standard error of the forecast *and* the critical *t*-value, not simply the standard error of the forecast.

35. B is correct. The coefficient of determination is 102.9152 ÷ 105.1303 = 0.9789.

36. A is correct. The standard error is the square root of the mean square error, or $\sqrt{0.0692} = 0.2631$.

37. B is correct. The *p*-value corresponding to the slope is less than 0.01, so we reject the null hypothesis of a zero slope, concluding that the fixed asset turnover explains the natural log of the net profit margin.

38. C is correct. The predicted natural log of the net profit margin is 0.5987 + (2 × 0.2951) = 1.1889. The predicted net profit margin is $e^{1.1889} = 3.2835\%$.

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PRACTICE PROBLEMS

1. A characteristic of Big Data is that:
   A. it involves formats with diverse structures.
   B. one of its traditional sources is business processes.
   C. real-time communication of it is uncommon due to vast content.

2. Which of the following statements is true in the use of ML:
   A. some techniques are termed "black box" due to data biases.
   B. human judgment is not needed because algorithms continuously learn from data.
   C. training data can be learned too precisely, resulting in inaccurate predictions when used with different datasets.

3. Text analytics is appropriate for application to:
   A. large, structured datasets.
   B. public but not private information.
   C. identifying possible short-term indicators of coming trends.

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SOLUTIONS

1. A is correct. Big Data is collected from many different sources and is in a variety of formats, including structured data (e.g., SQL tables), semistructured data (e.g., HTML code), and unstructured data (e.g., video messages).

2. C is correct. Overfitting occurs when the ML model learns the input and target dataset too precisely. In this case, the model has been "overtrained" on the data and is treating noise in the data as true parameters. An ML model that has been overfitted is not able to accurately predict outcomes using a different dataset and might be too complex.

3. C is correct. Through the text analytics application of NLP, models using NLP analysis might incorporate non-traditional information to evaluate what people are saying—through their preferences, opinions, likes, or dislikes— in an attempt to identify trends and short-term indicators—for example, about a company, a stock, or an economic event—to forecast coming trends that may affect investment performance in the future.